KAIZEN MATHS

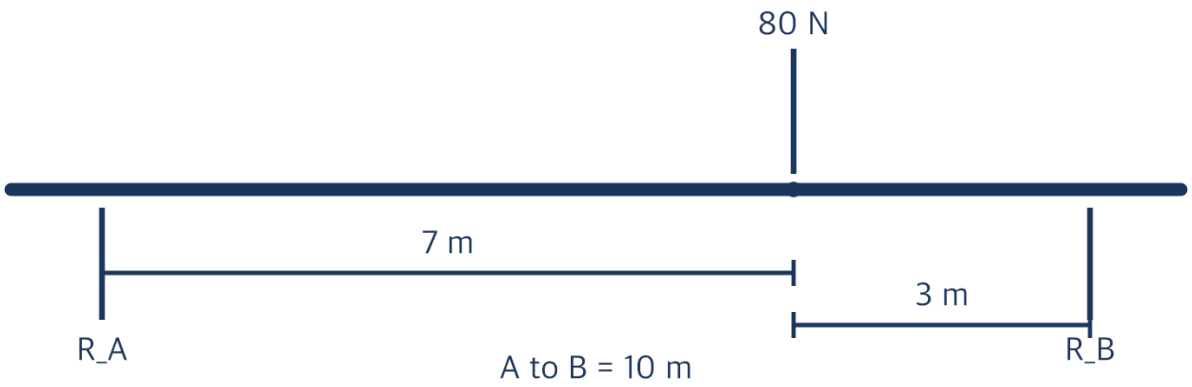
# Lesson starter

**Moments and Equilibrium**

Reactions on Beams

Name _____

SA1




A light beam AB is supported at A and B. A point load of 80 N acts 7 m from A. Find the reactions at A and B.



Show your working

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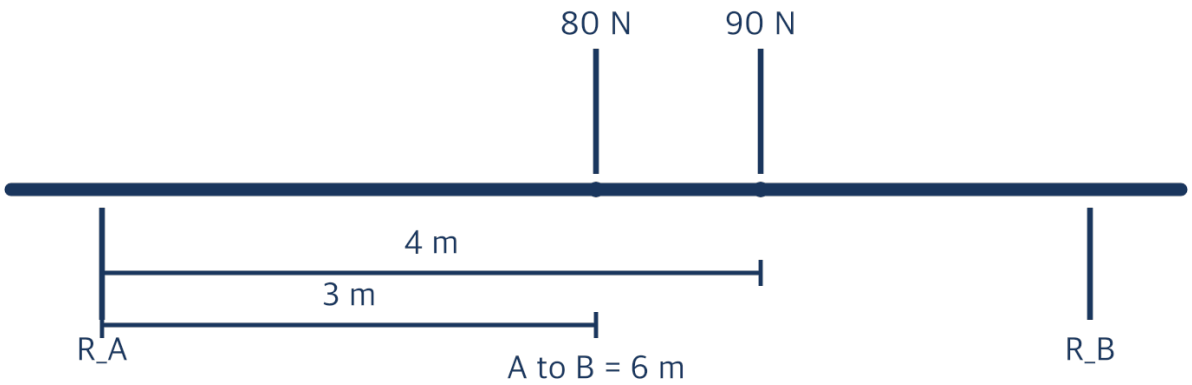
Lesson starter

Moments and Equilibrium

Reactions on Beams

Name _____

SB1




A uniform beam AB has weight 80 N and length 6 m. It is supported at A and B. A point load of 90 N acts 4 m from A. Find the reactions at A and B.



Show your working

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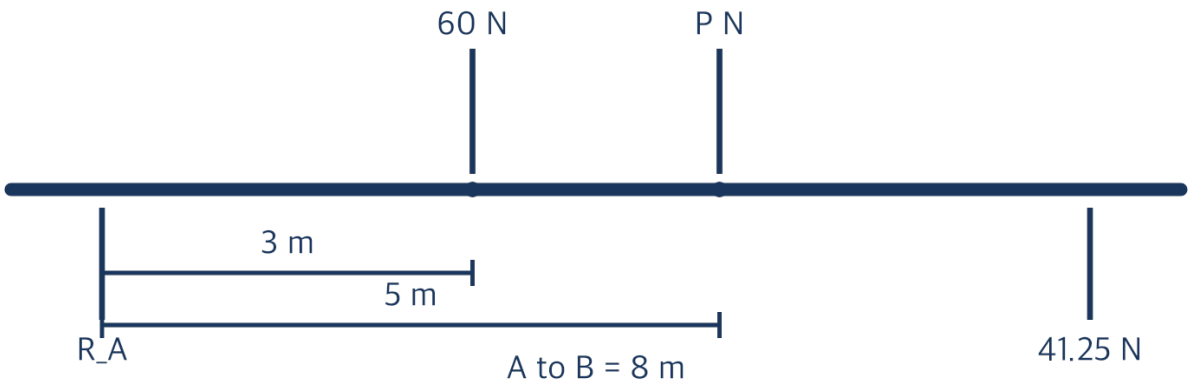
# Lesson starter

**Moments and Equilibrium**

Reactions on Beams

Name _____

SC1




A light beam AB is supported at A and B. The reaction at B is 41.25 N. Find the unknown load P.



Show your working

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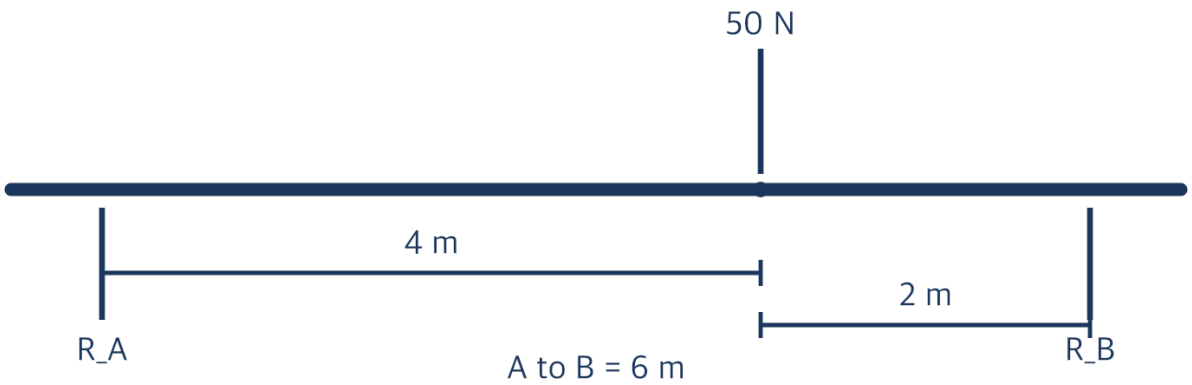
Lesson starter

**Moments and Equilibrium**

Reactions on Beams

Name _____

SD1




A light beam AB is supported at A and B. A point load of 50 N acts 4 m from A. Find the reactions at A and B.



Show your working

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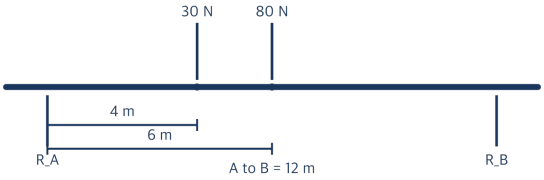
# Exit ticket

**Moments and Equilibrium**

Reactions on Beams

Name _____

**EA1**




A uniform beam AB has weight 80 N and length 12 m. It is supported at A and B. A point load of 30 N acts 4 m from A. Find the reactions at A and B.

MEC-4D51-L3 · EA

Show your working

KAIZEN MATHS

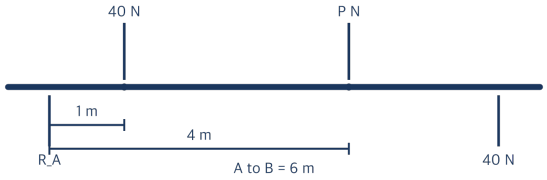
# Exit ticket

**Moments and Equilibrium**

Reactions on Beams

Name _____

**EB1**




A light beam AB is supported at A and B. The reaction at B is 40 N. Find the unknown load P.

MEC-4D51-L3 · EB

Show your working

KAIZEN MATHS

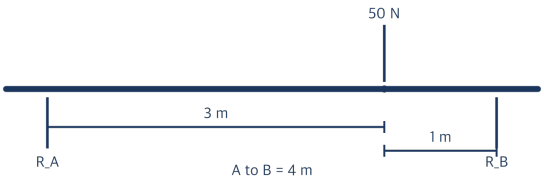
# Exit ticket

**Moments and Equilibrium**

Reactions on Beams

Name _____

**EC1**




A light beam AB is supported at A and B. A point load of 50 N acts 3 m from A. Find the reactions at A and B.

MEC-4D51-L3 · EC

Show your working

KAIZEN MATHS

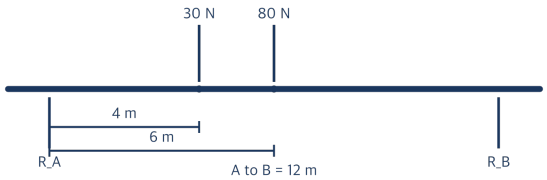
# Exit ticket

**Moments and Equilibrium**

Reactions on Beams

Name _____

**EA1**




A uniform beam AB has weight 80 N and length 12 m. It is supported at A and B. A point load of 30 N acts 4 m from A. Find the reactions at A and B.

MEC-4D51-L3 · EA

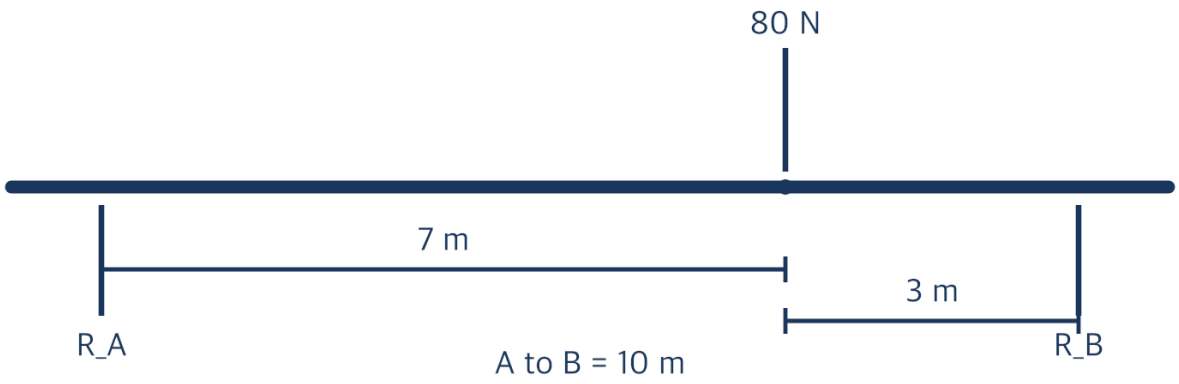
Show your working

# Teacher answers & working

**Moments and Equilibrium · Reactions on Beams**

MEC-4D51-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

**SA1**




A light beam AB is supported at A and B. A point load of 80 N acts 7 m from A. Find the reactions at A and B.

**Answer:**

$R_A = 24$ N, $R_B = 56$ N

**Working:**

Take moments about A to find R_B.

$R_B \times 10 = 80 \times 7$

$R_B = 560 \div 10 = 56$ N

Use vertical equilibrium.

$R_A + R_B = 80$

$R_A = 80 - 56 = 24$ N

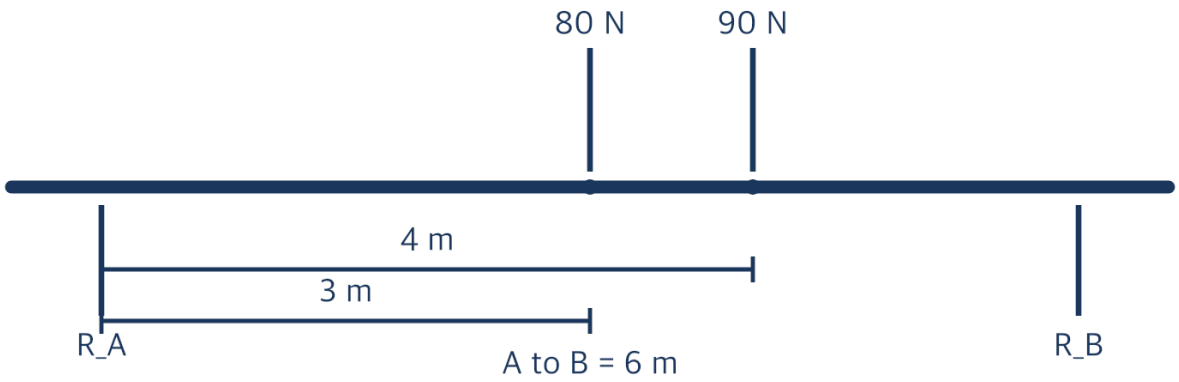
$R_A = 24$ N and $R_B = 56$ N.

# Teacher answers & working

**Moments and Equilibrium · Reactions on Beams**

MEC-4D51-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

**SB1**




A uniform beam AB has weight 80 N and length 6 m. It is supported at A and B. A point load of 90 N acts 4 m from A. Find the reactions at A and B.

**Answer:**

$R_A = 70$ N, $R_B = 100$ N

**Working:**

The weight of a uniform beam acts at its centre.

The beam's weight acts 3 m from A.

Take moments about A to find R_B.

$R_B \times 6 = 80 \times 3 + 90 \times 4$

$R_B \times 6 = 240 + 360 = 600$

$R_B = 600 \div 6 = 100$ N

Use vertical equilibrium.

$R_A + R_B = 80 + 90 = 170$

$R_A = 170 - 100 = 70$ N

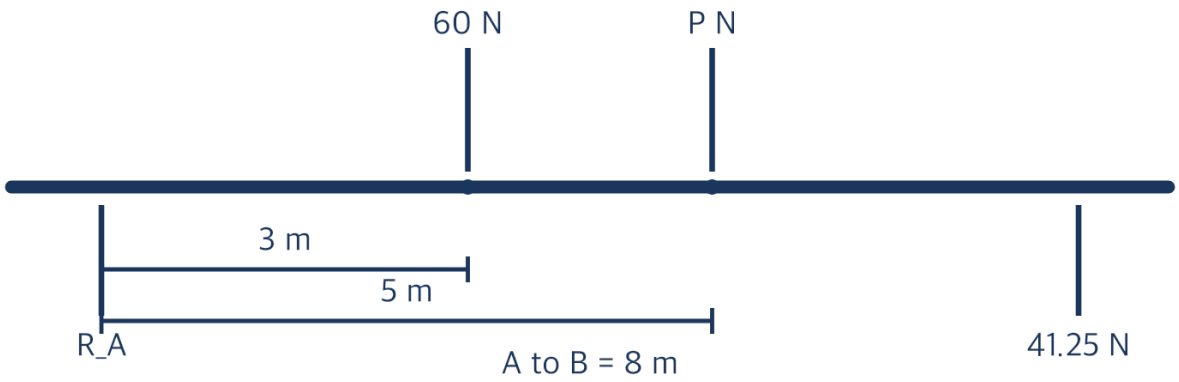
$R_A = 70$ N and $R_B = 100$ N.

# Teacher answers & working

**Moments and Equilibrium · Reactions on Beams**

MEC-4D51-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

**SC1**




A light beam AB is supported at A and B. The reaction at B is 41.25 N. Find the unknown load P.

**Answer:**

$P = 30$ N

**Working:**

Take moments about A.

$R_B \times 8 = 60 \times 3 + P \times 5$

Substitute the known reaction at B.

$41.25 \times 8 = 180 + 5P$

$330 = 180 + 5P$

Solve for P.

$5P = 330 - 180 = 150$

$P = 150 \div 5 = 30$

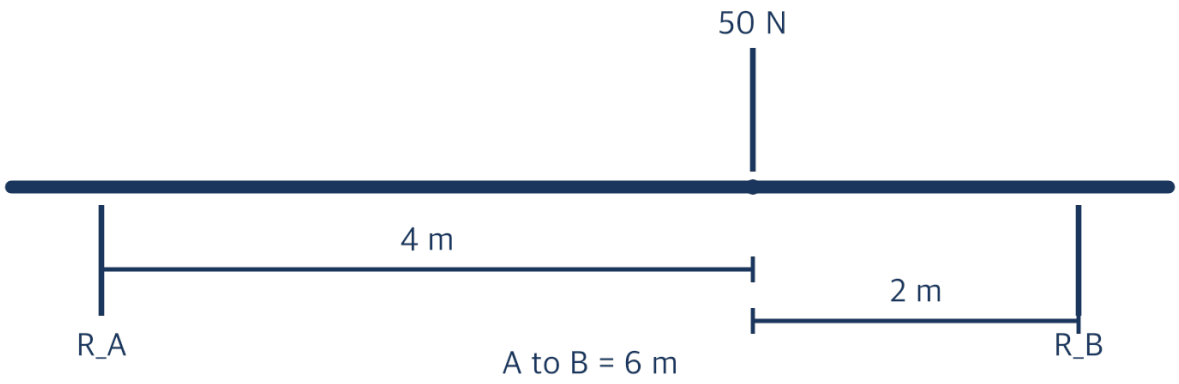
$P = 30$ N.

# Teacher answers & working

**Moments and Equilibrium · Reactions on Beams**

MEC-4D51-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

**SD1**




A light beam AB is supported at A and B. A point load of 50 N acts 4 m from A. Find the reactions at A and B.

**Answer:**

$R_A = 16.67$ N, $R_B = 33.33$ N

**Working:**

Take moments about A to find R_B.

$R_B \times 6 = 50 \times 4$

$R_B = 200 \div 6 = 33.33$ N

Use vertical equilibrium.

$R_A + R_B = 50$

$R_A = 50 - 33.33 = 16.67$ N

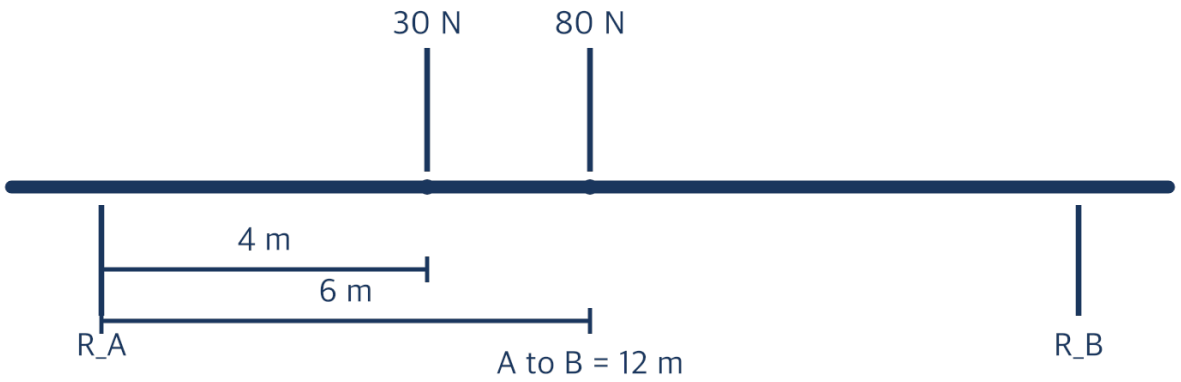
$R_A = 16.67$ N and $R_B = 33.33$ N.

# Teacher answers & working

**Moments and Equilibrium · Reactions on Beams**

MEC-4D51-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EA1




A uniform beam AB has weight 80 N and length 12 m. It is supported at A and B. A point load of 30 N acts 4 m from A. Find the reactions at A and B.

**Answer:**

$R_A = 60$ N, $R_B = 50$ N

**Working:**

The weight of a uniform beam acts at its centre.

The beam's weight acts 6 m from A.

Take moments about A to find R_B.

$R_B \times 12 = 80 \times 6 + 30 \times 4$

$R_B \times 12 = 480 + 120 = 600$

$R_B = 600 \div 12 = 50$ N

Use vertical equilibrium.

$R_A + R_B = 80 + 30 = 110$

$R_A = 110 - 50 = 60$ N

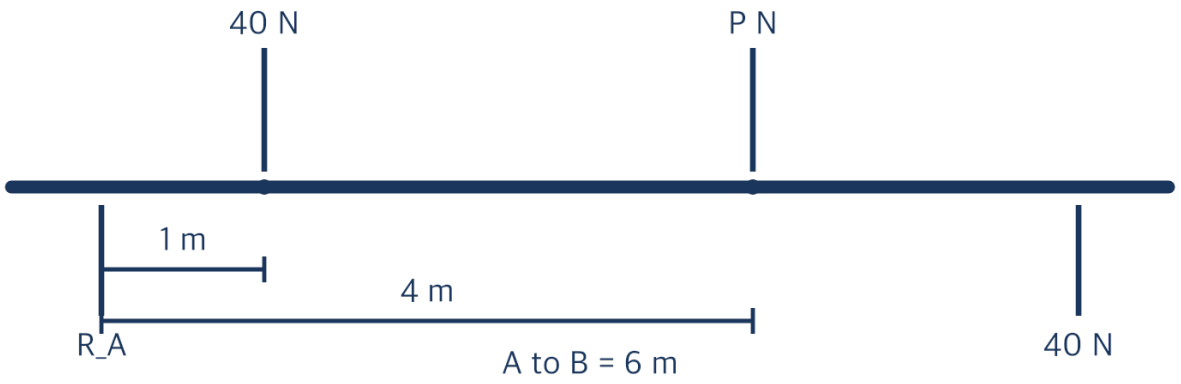
$R_A = 60$ N and $R_B = 50$ N.

# Teacher answers & working

**Moments and Equilibrium · Reactions on Beams**

MEC-4D51-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EB1




A light beam AB is supported at A and B. The reaction at B is 40 N. Find the unknown load P.

**Answer:**

$P = 50$ N

**Working:**

Take moments about A.

$R_B \times 6 = 40 \times 1 + P \times 4$

Substitute the known reaction at B.

$40 \times 6 = 40 + 4P$

$240 = 40 + 4P$

Solve for P.

$4P = 240 - 40 = 200$

$P = 200 \div 4 = 50$

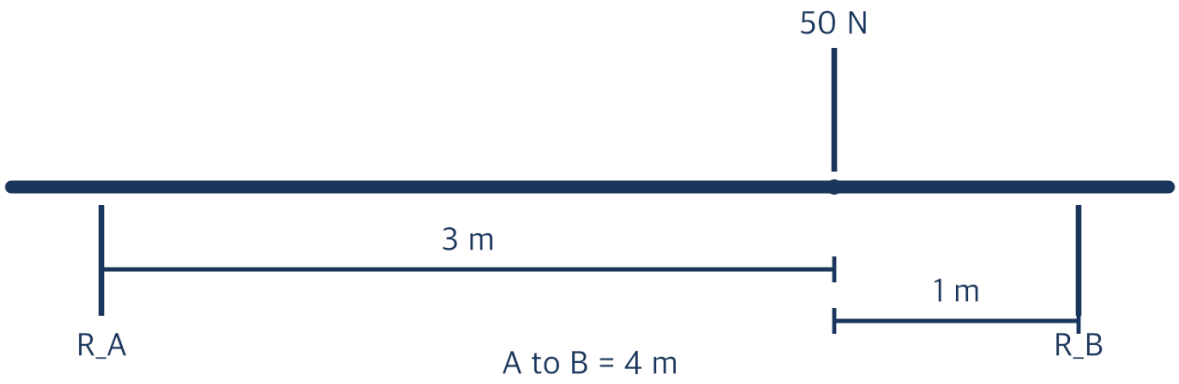
$P = 50$ N.

# Teacher answers & working

**Moments and Equilibrium · Reactions on Beams**

MEC-4D51-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## EC1




A light beam AB is supported at A and B. A point load of 50 N acts 3 m from A. Find the reactions at A and B.

**Answer:**

$R_A = 12.5$ N, $R_B = 37.5$ N

**Working:**

Take moments about A to find R_B.

$R_B \times 4 = 50 \times 3$

$R_B = 150 \div 4 = 37.5$ N

Use vertical equilibrium.

$R_A + R_B = 50$

$R_A = 50 - 37.5 = 12.5$ N

$R_A = 12.5$ N and $R_B = 37.5$ N.